

Faculty of Computers and Data Science

DS Tools · Spring 2026 · Final Project

Find The Story Hidden in Your Data

E-Commerce User Behavior — Purchase Prediction

Submitted by

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May 2026

1. Understanding the Data

The dataset used is **ecommerce_user_behavior.csv**, imported via PROC IMPORT. Metadata was inspected with PROC CONTENTS and the first 10 observations were printed with PROC PRINT. A separate explanation file (*data_explanation.csv*) was also imported and printed to describe each column.

Problem, Goal, and Explanation

As written in the code:

```
what we know from the DATA
the problem: classification
our goal: predict whether the customer will buy from the site or not
explanation: it affects sales and the profit of the company
```

Descriptive Statistics

PROC MEANS was applied to the full dataset to obtain summary statistics for all numeric variables. PROC SUMMARY with NMISS counted missing values in numeric columns. PROC FREQ with MISSING checked missing counts for: device_type, gender, discount_seen, ad_clicked, returning_user, and purchase.

2. Investigating the Data

Six visualizations were produced on the raw dataset using PROC SGPLOT to explore distributions, outliers, and relationships between variables.

#	Chart Type	Variables	Purpose
1	Histogram + Normal Density	time_on_site	Identify distribution of session time
2	Horizontal Boxplot	time_on_site	Identify outliers
3	Clustered Bar Chart	gender vs time_on_site grouped by device_type	Device usage by gender
4	Scatter Plot	time_on_site vs pages_viewed coloured by purchase	Buying depth vs session length
5	Clustered Bar Chart	device_type vs pages_viewed grouped by retailer	Pages viewed by device and loyalty
6	Scatter Plot (Loyalty Matrix)	previous_purchases vs avg_session_time coloured by purchase	Used by purchase engagement

3. Fixing the Data

Dropping Useless Columns

The column `user_id` was dropped as it carries no predictive value.

```
DATA ECB_cleaned; SET ECB; drop user_id; RUN;
```

Handling Missing Values — Categorical (PROC SQL)

- `device_type` — filled with 'Mobile'
- `gender` — filled with 'Male'; also corrected lowercase 'female' to 'Male'
- `discount_seen` — filled with 1
- `ad_clicked` — filled with 0
- `returning_user` — filled with 1

Handling Missing Values — Numerical

Column averages were computed into macro variables using `SELECT INTO`, then used to impute nulls in: `age`, `pages_viewed`, `previous_purchases`, `cart_items`, `avg_session_time`, `bounce_rate`, and `time_on_site`.

Standardization

`PROC STANDARD (MEAN=0, STD=1)` was applied to: `pages_viewed`, `previous_purchases`, `cart_items`, `bounce_rate`, `time_on_site`, `avg_session_time`.

One-Hot Encoding

`PROC GLMMOD` encoded `device_type` and `gender` into binary columns: Desktop, Mobile, Tablet, Female, Male. These were joined to the cleaned dataset via `PROC SQL` inner join to produce `ECB_model`.

After-Cleaning Verification

`PROC SUMMARY NMISS` was run on `ECB_model` to confirm zero missing values. `PROC PRINT (OBS=10)` displayed the cleaned data. A post-cleaning histogram of `time_on_site` confirmed standardization (visualization 7).

Correlation Heatmap — Pre-Feature Engineering (Visualization 8)

`PROC CORR` was run on `ECB_cleaned` across all numeric features including purchase. Results were reshaped to long format and plotted as an 850x850 px heatmap using `PROC SGPLOT HEATMAPPARM`, coloured blue (negative) through white (zero) to red (positive).

4. Creating New Insights — Feature Engineering

Four new cubic features were created to capture non-linear relationships and help the model better separate purchasers from non-purchasers:

New Feature	Formula	Stated Reason
time_on_site_CUBIED	time_on_site ** 3	Catch non-linear effect of session duration
pages_viewed_CUBIED	pages_viewed ** 3	Catch non-linear browsing depth signal
bounce_rate_CUBIED	bounce_rate ** 3	Amplify strong bounce signals at extremes
cart_items_CUBIED	cart_items ** 3	Catch non-linear cart size / purchase intent

A second correlation heatmap (750x750 px, visualization 9) was produced on ECB_model after feature engineering to evaluate the new features' relationships with the target variable purchase.

5. Building the Model

Train / Test Split

The dataset was split by row index (VAR1): records with VAR1 < 6000 formed the training set; records with VAR1 >= 6000 formed the test set.

Feature Selection — Impact on Purchase

```
time_on_site : the longer the time the more likely to buy
pages_viewed : the more pages the more likely to buy
previous_purchases: the more times the customer purchases the more loyal and more likely
to buy
cart_items : the more items the more likely to buy
bounce_rate : the less it is, the more likely the customer stays and buys
cubic features : to catch non-linear effects
```

PROC LOGISTIC — Training and Scoring

A logistic regression model was trained using PROC LOGISTIC with DESCENDING option (event = '1'). The SCORE statement was used to score both the training and test sets. The trained model was stored with the STORE statement.

```
PROC LOGISTIC DATA=train DESCENDING NOPRINT;
MODEL purchase(event="1") = time_on_site pages_viewed pages_viewed_CUBIED
previous_purchases cart_items bounce_rate
time_on_site_CUBIED bounce_rate_CUBIED cart_items_CUBIED;
SCORE DATA=train OUT=training_scores;
SCORE DATA=test OUT=testing_scores;
store work.my_model;
RUN;
```

6. Model Evaluation

Predicted labels (I_purchase) and actual labels (F_purchase) were extracted from the score output tables into TRAIN_RESULTS and TEST_RESULTS via PROC SQL. A binary correct flag was computed in a DATA step (correct = 1 if prediction matches actual). PROC MEANS computed the mean of the correct flag to yield training accuracy and test accuracy.

```
DATA TRAIN_CHECK; SET TRAIN_RESULTS; correct = (purchase = predict); RUN;
DATA TEST_CHECK; SET TEST_RESULTS; correct = (purchase = predict); RUN;
PROC MEANS DATA=TRAIN_CHECK MEAN; VAR correct; TITLE 'TRAINING ACCURACY'; RUN;
PROC MEANS DATA=TEST_CHECK MEAN; VAR correct; TITLE 'TEST ACCURACY'; RUN;
```

7. Making It Useful — Real-World Application

As written in the code:

```
by using the chosen features we can decide whether or not the customer is going
to buy from the site or won't, which we can use later to do more things for the
customers who will buy from the site, like giving them more credits, cash back,
and gifts; furthermore...
```


Requirements Checklist

#	Requirement	Status
1	At least 1 classification model	
2	At least 4 features after engineering	
3	Show before / after cleaning	
4	At least 8 visualizations	
5	PROC IMPORT used	
6	PROC MEANS used	
7	PROC LOGISTIC used	
8	Task 1 — Understand the Data	
9	Task 2 — Investigate the Data	
10	Task 3 — Fix the Data	
11	Task 4 — Create New Insights	
12	Task 5 — Build a Model	
13	Task 6 — Explain the Model	
14	Task 7 — Make It Useful	